HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed embedded C code for microcontrollers, focusing on **CPU architecture** and real-time constraints within an RTOS environment, contributing to a **15% reduction** in system boot time.
- Implemented **memory management** strategies in C for resource-constrained embedded systems, optimizing data structures to improve execution efficiency by **20%**.
- Debugged complex kernel-level issues using **JTAG and T32 debuggers**, tracing execution flow across inter-processor communication channels to resolve race conditions.
- Designed and implemented **inter-processor communication protocols** for multi-core CPU systems, ensuring reliable data exchange between different processing units.
- Wrote and maintained **Windows Kernel-Mode Driver Framework (KMDF)** components for device interaction, ensuring stable operation within the Windows OS.
- Contributed to **software design and development** for embedded systems, participating in code reviews and adhering to strict coding standards to enhance maintainability.
- Containerized C/C++ based embedded applications using **Docker** for consistent build and test environments on **Linux**, streamlining the development workflow.

SKILLS

RTOS – Zephyr, FreeRTOS, ThreadX

Programming Languages – C, C++, Python, SQL, Java

Embedded Systems – CPU Architecture, Kernel Development, Memory Management, Inter-processor Communication, Embedded Debugging

Tools & Debugging – JTAG, T32, GDB, Jenkins, Git, Jira, Confluence

Operating Systems – Linux, Windows (Kernel-Mode Driver Framework - KMDF)

Containerization – Docker

PROJECTS

Embedded C Development for Real-Time Monitoring System

- Developed real-time data acquisition modules in **C** for an embedded monitoring system, processing sensor inputs at **~500Hz**.
- Implemented **low-level drivers** for custom hardware interfaces, ensuring reliable data capture and transmission within a **FreeRTOS** environment.
- Optimized critical code paths using embedded debugging tools to achieve **p95 latency** below **2ms** for core processing loops.

RTOS Kernel Analysis and Optimization

- Analyzed and profiled the performance of a **Zephyr RTOS** kernel on an ARM Cortex-M microcontroller, identifying bottlenecks in task scheduling.
- Refactored memory allocation routines and inter-task communication mechanisms, leading to a **18% improvement** in system responsiveness.
- Documented findings and proposed optimizations for kernel behavior, contributing to improved system stability and resource utilization.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)